Brainstorming and Idea Prioritization

Phase 1: Brainstorming & Ideation

1. Purpose of Brainstorming

The purpose of brainstorming was to identify a practical problem that could be addressed through a web application and Generative AI. Travel planning was selected because a traveller typically has to combine destination, dates, budget, group size, interests and travel style before a useful itinerary can be prepared. These details are often collected from different websites and then arranged manually into a day-wise schedule.

The project idea was evaluated not only by novelty but also by feasibility. A student project needs a clear user journey, an achievable implementation scope, visible results that can be demonstrated, and a technical design that can be explained from source code. Trip Genius AI provides a defined input-output workflow and an opportunity to integrate an AI model with persistent storage.

2. Ideas Considered

AI-powered itinerary planner

This idea converts a structured set of travel preferences into a proposed daily plan. It offers a direct use for a generative model and allows the user to review, save, reopen and print the result. The idea was prioritized because it is useful as a planning assistant without requiring the project to implement booking transactions.

General destination information portal

A destination portal could present descriptions and attraction lists. However, without personalization, the user would still need to assemble the information into a schedule. This concept was therefore less aligned with the goal of generating a tailored itinerary.

Complete travel booking platform

A booking platform could connect transport, hotels, restaurants and payments. This would require provider contracts, verified inventory, payment security and transaction handling. Those requirements are beyond the current application, so they remain future scope rather than being claimed as implemented features.

Travel expense tracker

A standalone expense tracker would help users record spending but would not directly solve the initial problem of creating an itinerary. Budget-aware planning is instead included as an input to the AI-generated draft.

3. Prioritization Criteria

- User value: the idea should reduce repetitive planning effort.
- Feasibility: the feature set should be achievable within the available project timeline.
- AI relevance: the model should produce meaningful content rather than be added only for appearance.
- Demonstrability: the complete flow should be visible in a browser demonstration.
- Extensibility: the architecture should allow later integration of weather, live place data and booking services.

4. Selected Idea and Rationale

Trip Genius AI was selected as the strongest fit. It accepts structured travel inputs, requests a structured itinerary from a hosted model, validates the response, and stores trip information in Supabase. The frontend also includes saved-trip actions and map-search links for hotels and restaurants. These capabilities create a coherent project rather than a collection of unrelated AI features.

5. Scope Guardrails

The brainstorming outcome deliberately separates current capabilities from possible future improvements. The application creates an itinerary draft and external Google Maps search URLs; it does not book services or directly verify hotel prices, ratings, opening hours, availability or dietary preparation. This boundary keeps the project description accurate and helps reviewers understand what remains to be developed.